

Project Lifecycle: 8-Phase Report

AI-powered learning assistant — from ideation to demonstration

PHASE 1 Brainstorming & Ideation Phase

- Identified a real student problem: scattered notes, PDFs, textbooks, and handwritten material across multiple sources.
- Rahul's case (a second-year engineering student preparing for exams and placements) framed the need for a smarter study support system.
- Concept formed: an AI assistant that automates summarizing, revision material creation, and self-assessment.

PHASE 2 Requirement Analysis Phase

- Core problems identified: manual note preparation, repetitive flashcard/quiz creation, lack of personalized study plans, and no centralized secure storage.
- Functional requirements: JWT + RBAC based authentication, study material upload, AI-generated summaries, flashcards, quizzes, and study plans.
- Software requirements: Node.js (v16+), npm (v8+), Express.js, MongoDB, Postman, VS Code.
- Hardware requirements: Intel i5 (8th Gen) / AMD Ryzen 5 or higher, 8GB RAM minimum (16GB recommended), 1GB storage.

PHASE 3 Project Design Phase

- Modular RESTful architecture following the MVC (Model–View–Controller) pattern.
- Six linked entities defined: User, Study Material, Summary, Flashcard, Quiz, and Study Plan.
- Core components: Client (React.js), Express Server, JWT Auth Middleware, Route Handlers, AI Service Layer (Gemini API), Database Layer (MongoDB/Mongoose).
- User flow designed: register/login → dashboard → upload material → generate & save AI content.

PHASE 4 Project Planning Phase

- Work broken into 5 Epics: Project Architecture, Setup & Configuration, Backend Development, Database Configuration, and Project Execution.
- Roles and responsibilities defined early — Student (learner features) vs Administrator (system oversight).
- Dependencies, folder structure, and environment variables planned ahead of development.

PHASE 5 Project Development Phase

- Built using Node.js, Express.js, and MongoDB/Mongoose in a modular file structure.

- Core modules implemented: index.js (entry point), src/utils/db.js (DB connection), src/middleware/auth.js (JWT auth), src/middleware/upload.js (file uploads).
- AI integration via the Google Gemini 2.5 Flash API (src/utils/gemini.js) for generating summaries, flashcards, quizzes, and study plans.
- Dependencies installed: express, mongoose, bcryptjs, jsonwebtoken, cors, dotenv, multer, @google/genai, and nodemon (dev).

PHASE 6 Project Testing Phase

- All REST endpoints tested using Postman.
- Endpoints covered: /api/auth/register, /api/auth/login, /api/material/upload, /api/materials/:id/summarize, /api/ai/flashcards, /api/ai/quiz, /api/ai/study-plan.
- Verified correct request/response handling and secure, role-based access control for each route.

PHASE 7 Project Documentation Phase

- Comprehensive documentation prepared covering system description, scenario-based case study, and requirements.
- Included architecture diagrams, ER diagrams, Mongoose schema definitions, and a complete API endpoint reference.
- Content organized by Epics for clarity, maintainability, and future onboarding.

PHASE 8 Project Demonstration Phase

- Backend Development Demo — recorded walkthrough of core backend functionality.
- API Testing / Execution Demo — live Postman verification of all endpoints.
- Source code and demo videos shared via Google Drive for review and submission.